

Performance of Narrow-Band Internet of Things (NB-IoT) Based on Repetition of Downlink Physical Channel

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Abstract- Narrow-Band Internet of Things (NB-IoT) offers low power and wide coverage for IoT services which reuse existing Long Term Evolution (LTE) technology. Coverage enhancement at weak signal area may be achieved by channel repetition to improve signal quality. In this paper, the performance of NB-IoT is evaluated for three different modes of deployments by repeating the downlink physical channel. It is proved that the increase in the number of repetitions will give better performance of NB-IoT system. It also shows that the standalone and guard band modes perform better compared to in-band mode.

Index Terms- NB-IoT, Repetition, In-band, Guardband, Standalone, NPDSCH

I. INTRODUCTION

NB-IoT is a radio access technology that is defined from the existing Long-Term Evolution (LTE) technology by 3GPP in middle of 2016 [4] [12]. The idea of NB-IoT introduction is to fulfill the demand of markets related to the IoT needs. The main objectives of this technology are to be able to effectively handle huge number of connected devices, to lengthen the battery operated by lowering the power consumptions and improve the weak and cell edge coverage by signal repetition in physical layer [1]. Deployment of NB-IoT are based on the guideline that have been defined in 3GPP Release-13.

NB-IoT supports repetition of the signals in order to boost the signal power for low powered devices that are in extreme coverage condition [9]. In [15], number of repetitions depends on the coverage enhancement level required by a User Equipment (UE). Repetition can exist in physical downlink channel and physical uplink channel.

NB-IoT can be deployed in three different modes – inband, guardband and standalone [1] [6]. Fig. 1 shows the inband mode where NB-IoT is deployed within the LTE carrier. NB-IoT requires only one physical resource block (PRB) from LTE legacy. PRB can be fully utilized by LTE whenever there is no NB-IoT traffic present. This is due to the integration of structure

and spectrum usage defined in LTE and NB-IoT. To support full flexible design, two modes of inband operation are defined. The first mode is named Same-PCI mode where the NB-IoT carrier has identical parameters, which are the physical cell ID (PCI) and number of transmit antennas as the LTE. The other mode is defined as Different-PCI where it considers options of having different cell ID and number of transmit antennas [6].

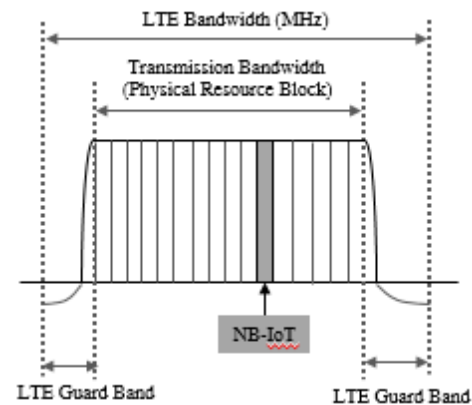


Fig. 1: Inband mode NB-IoT

Guardband mode means the deployment of NB-IoT in channel bandwidth that are reserved for the guardband of legacy LTE band as illustrated in Fig. 2. One of the benefits of guardband mode is none of PRB in legacy LTE is used. In addition, all symbols in subframe can be used for guardband operation compared to inband mode where there are few symbols will be used by LTE for control channel and reference signals [1] [6].

For standalone mode, the NB-IoT is deployed in non-LTE carrier (GSM carrier) as shown in Fig. 3. GSM carriers mainly operates in 900 MHz and 1800 MHz frequency bands (present spectrum in nowadays market). Deploying NB-IoT in these carriers basically an advantage as it can accelerate its readiness time to be in market [6].

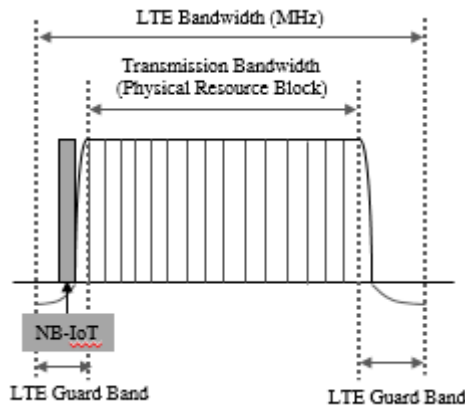


Fig. 2: NB-IoT Deployment in Guardband Mode

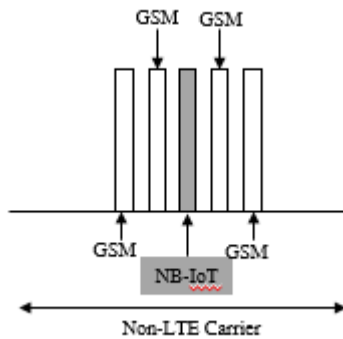


Fig. 3: NB-IoT Deployment in Standalone Mode

Since NB-IoT is part of LTE family, therefore there are few common characteristics of NB-IoT and LTE such as the length of radio frame and the PRB. In NB-IoT, the system bandwidth is fixed to 180 kHz spectrum which is equivalent to one PRB. Thus, the number of resource block in NB-IoT system bandwidth is always one [7] with 12 subcarriers separated by 15 kHz guard band. Fig. 4 shows the even and odd downlink frames. Each frame has a duration of 10 ms and consists of 10 subframes.

In 3GPP Rel-13, there are new channels and signals are derived based on LTE channels with some modification to fit into the narrow bandwidth used by downlink (DL) and uplink (UL) in NB-IoT. Narrow-band Primary Synchronization Signal (NPSS) and Narrow-band Secondary Synchronization Signal (NSSS) provide information on time and frequency synchronization. NPSS is transmitted in subframe 5 in all radio frames while NSSS is transmitted in subframe 9 in even radio frames only. Cell reference is identified by the Narrow-band Reference Signal (NRS) cell.

Three new downlink channels are Narrow-band Physical Broadcast Channel (NPBCH), Narrow-band Physical Downlink Shared Channel (NPDSCH) and Narrow-band Downlink Control Channel (NPDCCH). NPBCH carries the Master Information Block (MIB) and occupies subframe 0 in every frame. NPDCCH carries downlink control information. NPDSCH carries the data intended for the specific user, in addition to paging, system information and random access response.

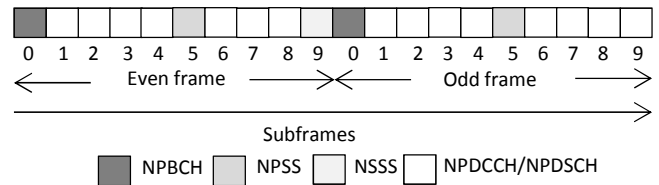


Fig. 4: Structure of NB-IoT downlink frame

In NB-IoT network, when User Equipment (UE) is in an ideal mode it will periodically monitor NPDCCH to check whether there is any paging message which indicates the presence of downlink traffic for the UE. Once the UE recognizes the presence of traffic for the UE, it will decode NPDSCH to identify the contents of the traffic. Therefore, successful decoding of NPDSCH is a crucial factor in providing a reliable NB-IoT service [6]. Thus, when the system enables the NPDSCH repetition transmission, it will increase the number of received signals detected and decoded at the weak signal area or cell edge. NB-IoT adjusts the number of transmission repetitions based on the signal strength and increases the number of transmission repetitions to ensure effective data transmission at coverage edges.

The NB-IoT signaling protocol uses new messages - Master Information Block NB (MIB-NB) and System Information Block Type I-NB (SIB1-NB). These two messages are different from the SIB and MIB messages in the LTE standard in terms of the content of the message, the scheduling type and the physical channel. Even though these two messages are different from the LTE standard, the eNodeB can still manage these new messages in the same manner by differentiating the UE types and messages [4]. MIB in legacy LTE only contains high-level information, but in MIB-NB it contains more detailed information to acquire SIB1-NB including the operation mode of NB-IoT. In [13], the specification of SIB1-NB messages is explained in detail. SIB1-NB messages are transmitted every 2560 ms, which is equivalent to 256 frames. Within this period, this message can be repeated several times. In the NB-IoT standard, the UE shall receive a minimum of one MIB-NB message and one SIB1-NB message before the Random Access Channel (RACH) process and connection establishment with the eNodeB.

II. SYSTEM MODEL

In order to ideally show the effect of repetition, the controlled channel performance has been considered. The performance of NPDSCH repetition will be presented by Block Error Rate (BLER) simulation under Additive White Gaussian Noise (AWGN). The improvement of signal quality is expected to be illustrated when the number of NPDSCH repetitions varies at three different modes of NB-IoT deployment - inband, guardband and standalone (non-LTE carrier). To have a comparable result, there are few constant variables configured in all three modes of NB-IoT deployment.

In [15] the number of NPDSCH repetitions and the transport block size (TBS) are affected depending on the NPDSCH channel whether it is carrying SIB1-NB or not. In this simulation, NPDSCH is configured to carry SIB1-NB. The number of repetitions depends on the TBS tables in [15].

The transport blocks of size 680 undergo 24-bit CRC encoding, convolutional encoding and rate matching stages in order to obtain the NPDSCH bits which are repeated according to repetition configured. Scrambling, QPSK modulation, layer mapping and precoding are applied to form complex NPDSCH symbols. AWGN is added to these NPDSCH symbols and then channel decoding and demodulation are performed to recover the transport block. After de-scrambling, the repetitive subframes are soft-combined before rate recover. The transport block error rate (BLER) of decoded blocks are calculated.

III. SIMULATION RESULTS

To present the effect of repetition, the channels are simulated as perfect channels where there is no channel propagation considered during simulation. Fig. 5 represents the effect of repetition transmission in inband mode. As NPDSCH channels are carrying SIB1-NB message, thus the number of repetitions can only be repeat either 4, 8 or 16 as defined in 3GPP standard. It shows for BLER 10^{-2} , the SNR at 4, 8 and 16 repetitions are -3.1 dB, -6.2 dB and -9.4 dB respectively. It clearly shows that by having repetition transmission, more signals can be detected correctly at low SNR as the number of repetitions is increasing. It shows that around 3 dB improvement can be achieved if the number of repetitions is doubled.

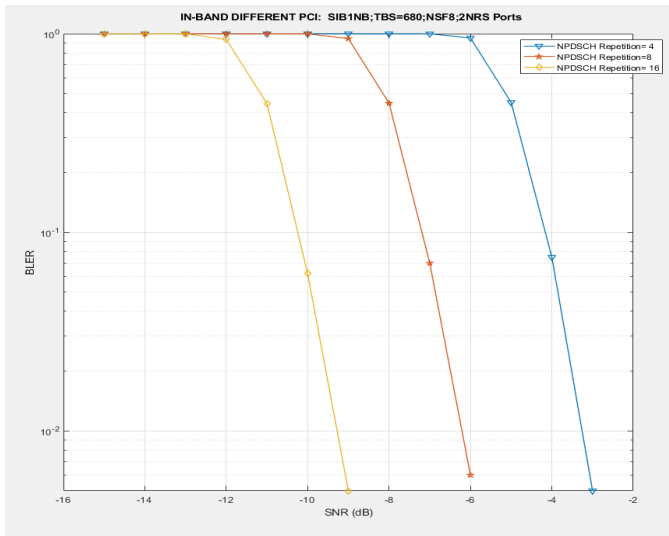


Fig. 5: NPDSCH Repetition in inband Mode

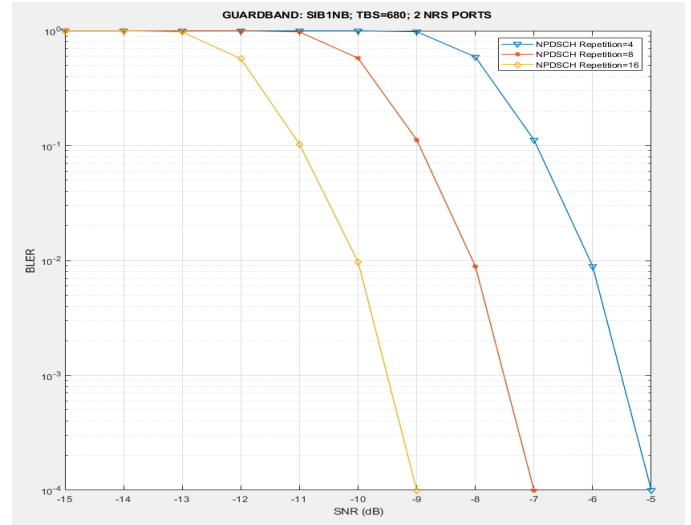


Fig. 6: NPDSCH Repetition in Guardband Mode

Similar to inband mode, the number of NPDSCH repetitions are varies at 4, 8 and 16. In Fig. 6, at BLER 10^{-2} , 4, 8 and 16 repetitions give SNR -6 dB, -8.1 dB and -10 dB respectively. For guardband and standalone modes no symbols are used for control and reference symbol. Due to this, more signal is detected and decoded at lower SNR compared to inband. It shows that around 2 dB improvement can be achieved if the number of repetitions is doubled.

Fig. 7 shows for BLER 10^{-2} , 4, 8 and 16 repetitions give SNR -6 dB, -8.1 dB and -10.2 dB respectively. Similar with other two modes, there are signal received at lower SNR which helps to improve the coverage improvement at cell edge or at weak signal area. It shows that around 2 dB improvement can be achieved if the number of repetitions is doubled.

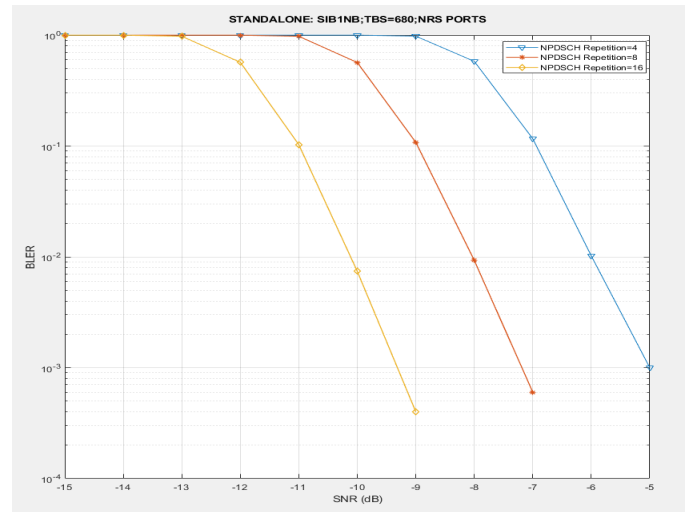


Fig. 7: NPDSCH Repetition in Standalone Mode

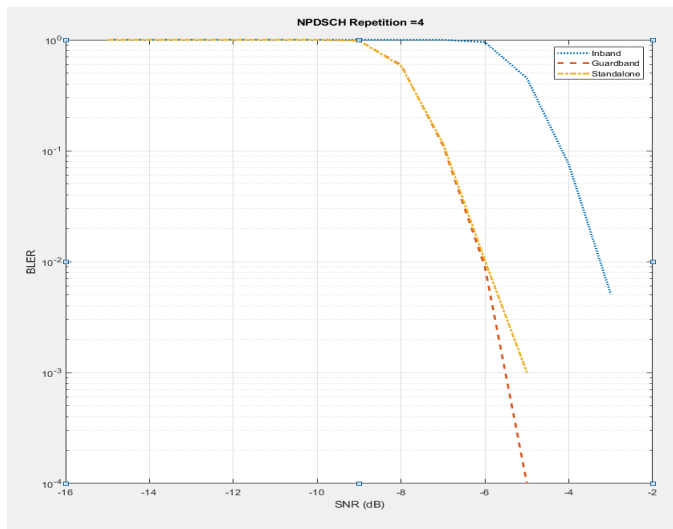


Fig. 8: NPDSCH Repetition = 4 for all three modes

Fig. 8 compares the effect of repetitions of NPDSCH channels when repeated four times in all three deployment modes. For guardband and standalone modes, the effect of repetition is the same for both modes. Differ from the other two modes, BLER 10^{-2} (1%), the SNR for inband is -3.2 dB while for other two modes the SNR is -6 dB. This result represented that, in guardband and standalone modes, the repetition able to boost the coverage better than in inband modes.

IV. CONCLUSION

Based on the result achieved, it can be concluded that repetition of NPDSCH improved signals quality at the lower SNR where more signals are detected and decoded correctly. As the number of NPDSCH channels repetition are increased the signal able to reach even at the lower SNR. Hence, it will enhance coverage at the weak area or cell edge. At lower SNR, more signals are detected and decoded in both guardband and standalone modes compared to inband mode. Regardless any modes of deployments, repetition transmission is possible to boost the coverage at lower SNR. Coverage improvement is important as NB-IoT users can have better experience in applying NB-IoT.

V. ACKNOWLEDGEMENT

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